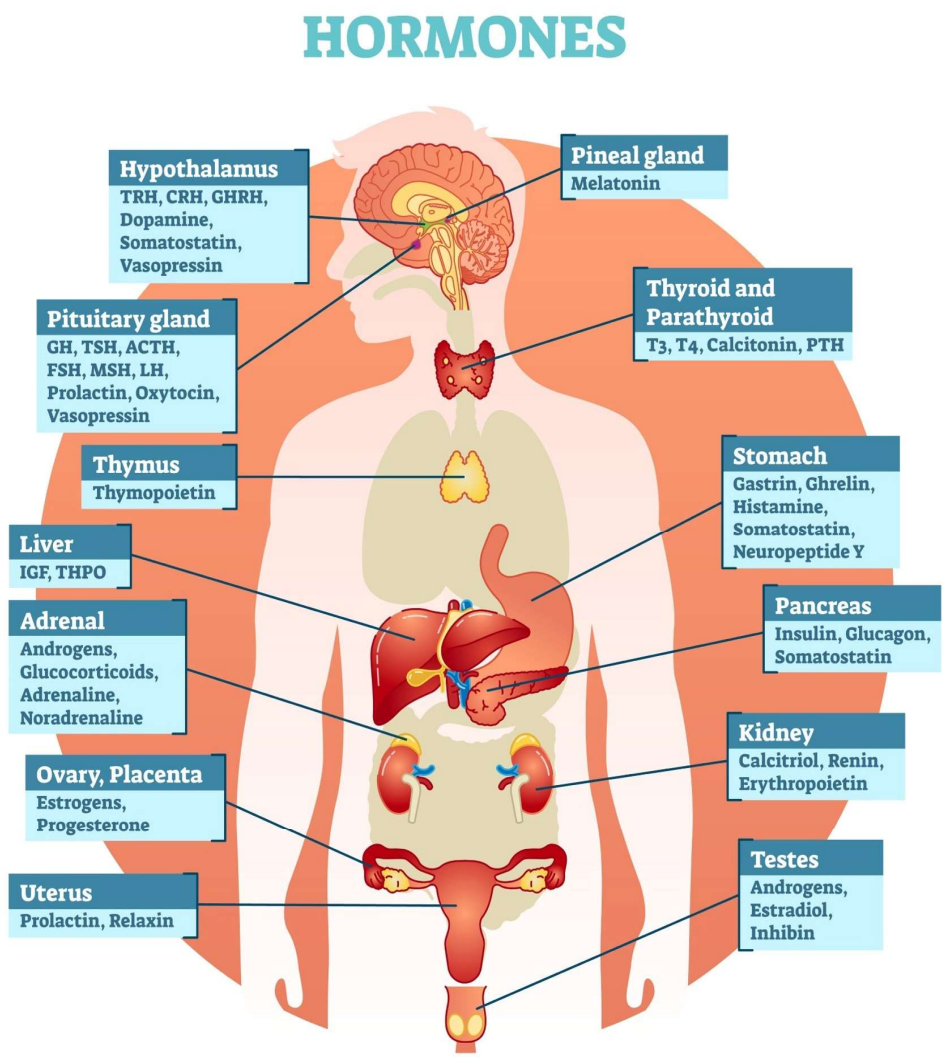
Physiology of Endocrine System (Endocrinology)

Lec1

By

staff Members of Physiology Department



- **What are hormones?**
- **Hormones are chemicals** that coordinate various functions in the body by carrying messages through blood to the target organs, skin , muscles and other tissues. These signals tell the body what to do and when to do it. Hormones are essential for life.
- Over 50 hormones were identified in the body till now.
- Hormones and the tissues (mainly glands) that create and release them make up the **endocrine system**.
- Integration of endocrine and nervous controls is a major function of the hypothalamus
- many different bodily processes are controlled via hormones, including:

- **Cell Metabolism.**
- **Homeostasis (constant internal balance), such as blood pressure and blood sugar regulation, fluid (water) and electrolyte balance and body temperature.**
- **Growth and development.**
- **cardiovascular performance.**
- **Gastrointestinal motility**
- **Renal function.**
- **Reproduction (Sexual function).**
- **Sleep-wake cycle.**
- **Mood.**
- **N.B) with hormones, a little bit goes a long way. So that, minor changes in levels can cause significant changes to your body and lead to certain conditions that require medical treatment:**

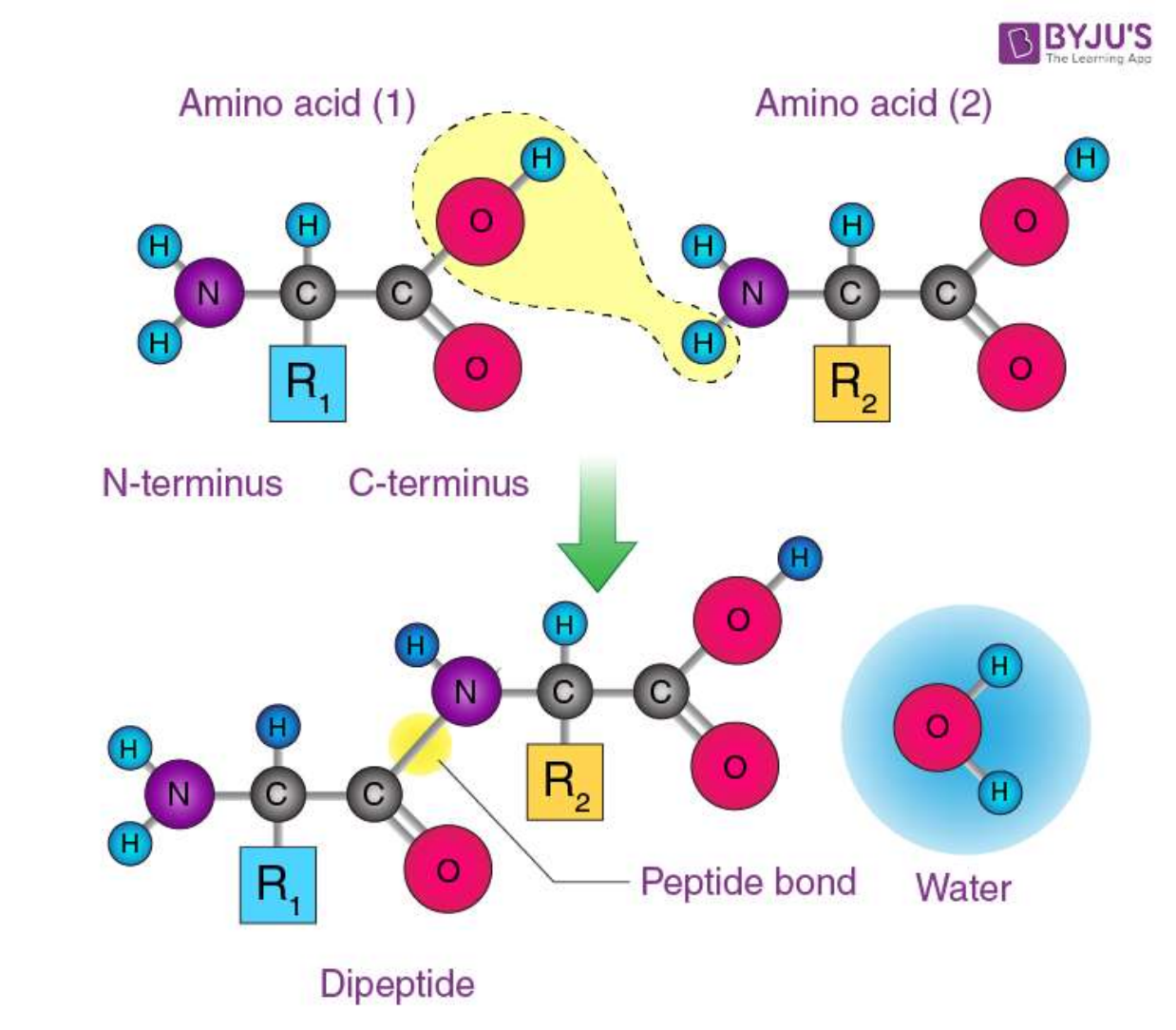
Hormones are grouped into three classes based on their structure

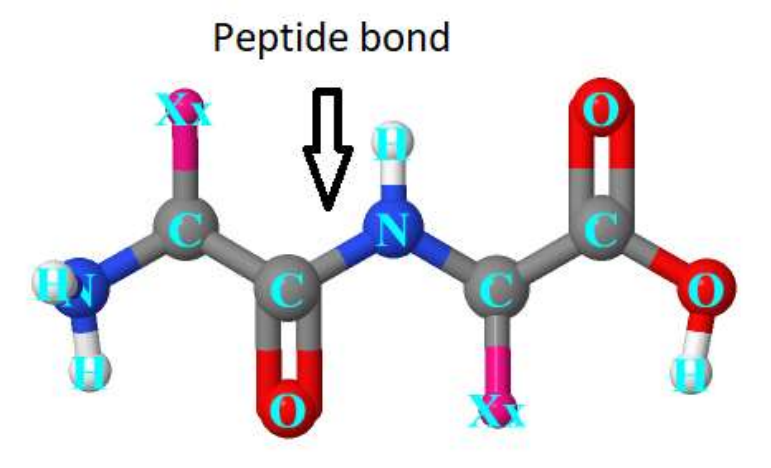
1. Protein/PolyPeptide Hormones (chains of amino acids):-

- **Most of body's hormone**
- **Made in rER then repackaged in Golgi app into vesicles then excreted from the cells.**
- **Water souble so, hard to cross cell membrane. Receptors on/in cell surface, after binding to R they initiate a cascade effect of 2nd messengers inside the cell.**
- **Soluble in saline, heat sensitive, Species specific & must be injected and form antibodies after repeated injections.**

A. Peptides **<200 amino acids (ADH, Oxytocin, Growth H.(GH), Insulin).**

B. Glycoproteins - **>200 amino acids** **m**ore complex protein hormones have carbohydrate (oligosaccharide sugar covalently bind o a.a.) side chains (**TSH, FSH, LH, HCG)**


BYJU'S
The Learning App
Amino acid (1)
Amino acid (2)
N-terminus
C-terminus
Peptide bond
Water
Dipeptide


Peptide bond

2. Lipid Derivatives:-

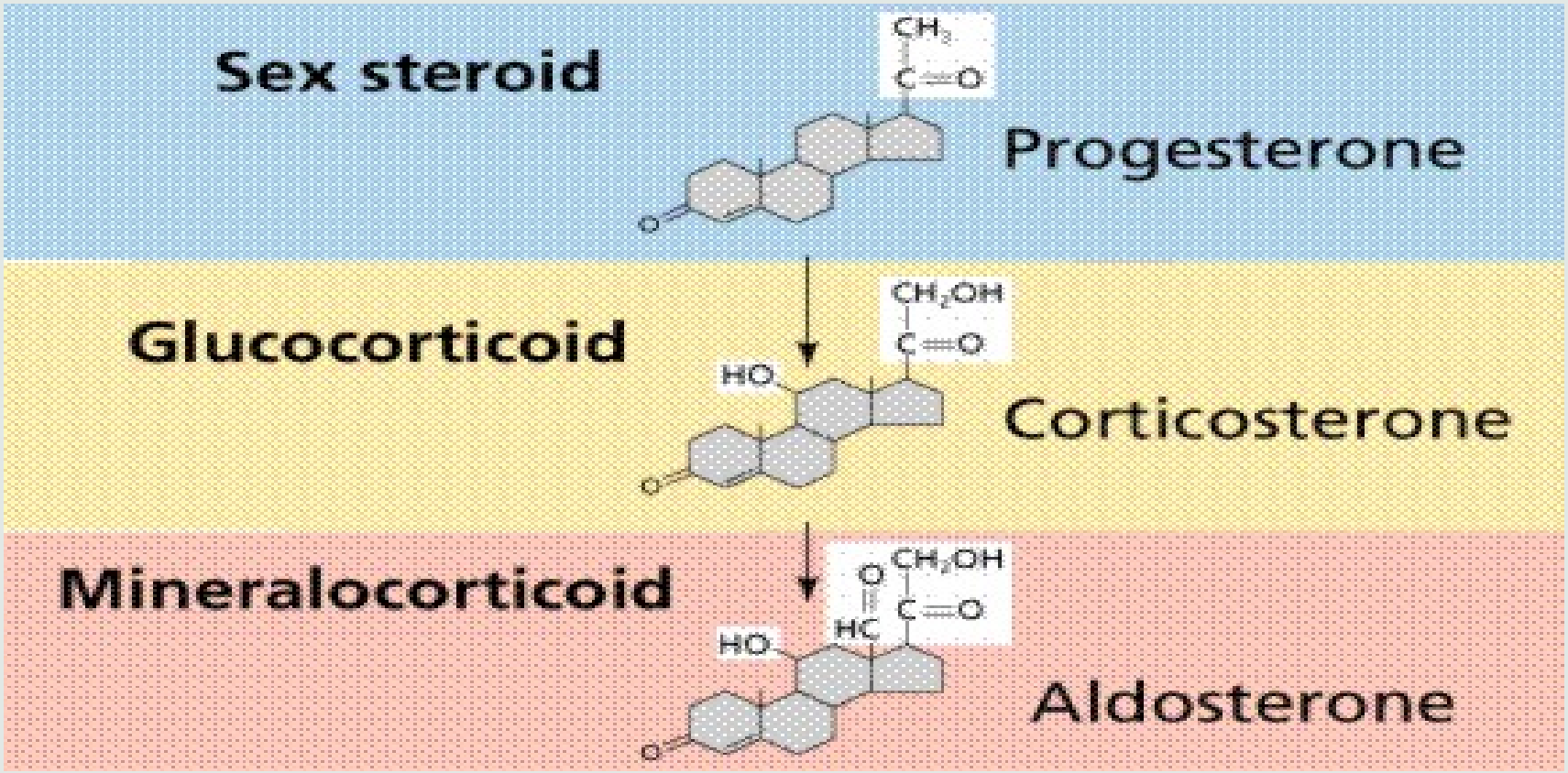
A. Steroid Hormones:- lipid H derived from cholesterol.

- **Receptors located inside the cell (nucleus or cytoplasm)**
- **Made up of lipids so easier to pass a cross the cell m, go inside the cell to signal the R as a primary messenger, and make change inside the cell via transcription and translation of new proteins and new products**

Steroid hormones are secreted from

1. **Gonads (Androgens, Estrogens, progesterone)**
2. **Adrenal cortex (glucocorticoide as cortisol, mineralocorticoid as aldosterone)**
3. **Placenta (Estrogens, progesterone).**
4. **Skin [vitamin D3 (cholecalciferol) and its active metabolite (calcitriol)].**

Structure of some steroid hormones and their pathways of formation.



B. Eicosanoids:-

- potent lipid signaling molecules (bioactive fats act as chemical messenger) derived from arachidonic acid (long chain fatty acids)
- Acting as "local hormones" that regulate vital functions such as inflammation, pain, fever, blood pressure, and immunity, muscle tone.
- Synthesized and broken down rapidly,
- Not circulate, they act near their origin (autocrine/paracrine), exert their local effects through interactions with cell surface or nuclear membrane receptors
- targeted by common drugs like NSAIDs (ibuprofen, aspirin) to manage symptoms like pain and inflammation (inhibit PGs & Thx but increase leukotriene).

1. Leukotrienes: secreted from leukocytes(coordinate inflammation)
2. Prostaglandins: inflammation, fever, pain, blood flow, and labor.
3. Thromboxane: Primarily involved in blood clotting (platelet aggregation).

3. Amino Acid Derivatives:- Why these H are special and get their own major class?

because of

1) As in this class, hormones are made up via manipulation of the single a.a. (tyrosine or tryptophan)

2) Hormones made up of one a.a. sometimes able to act like Polypeptide & protein H and some times act like steroids.

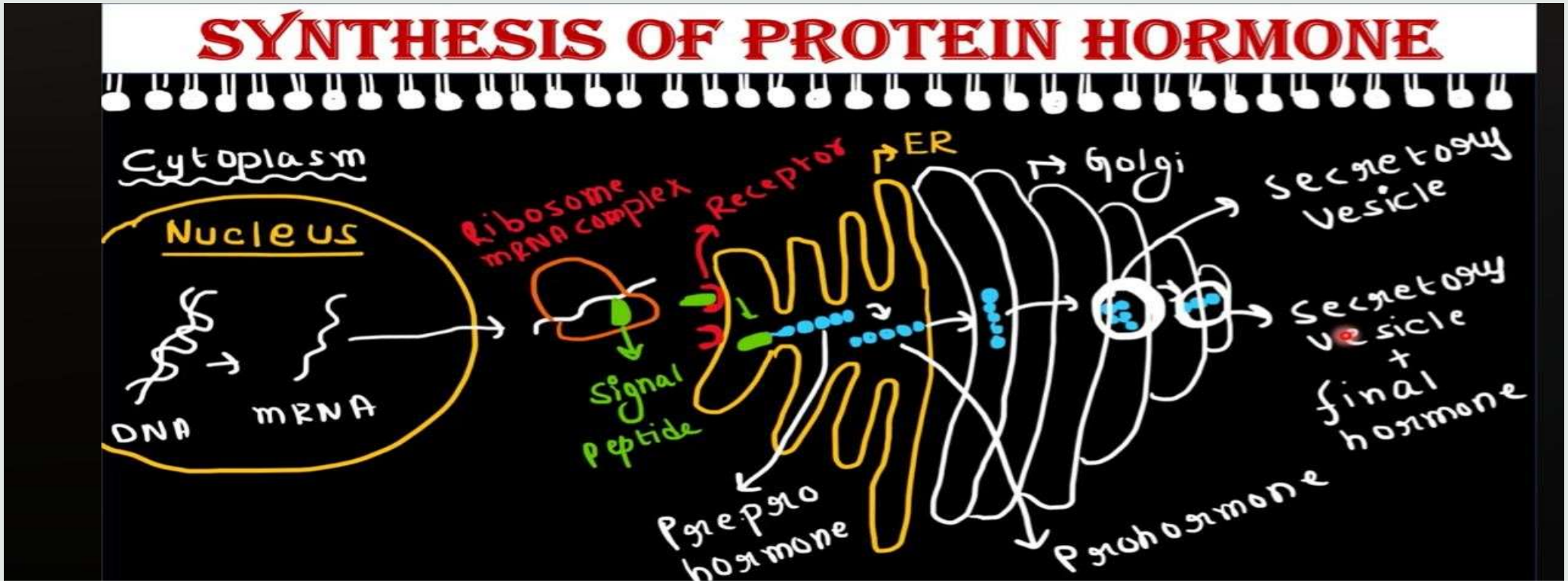
a. Tyrosine derivatives

1. **Thyroid hormones (T3-T4)** act similarly like steroids (binding to R inside the cell)
2. **Catecholamines (epinephrine, nor-epinephrine, dopamine)** act like protein & polypeptides H by binding to cell m R & release 2nd messenger inside the cell

b. Tryptophan derivative (melatonin H).

Synthesis, Storage and Secretion:-

1. Protein H: are synthesized as large precursor molecules (preprohormones) in the rough endoplasmic reticulum, processed in the Golgi apparatus, and stored in secretory granules. Upon stimulation, these vesicles are released into the bloodstream via exocytosis. They are generally water-soluble, allowing them to travel freely in the plasma. These vesicles are



Synthesis, Storage, and Secretion:

2. Steroid hormones: are synthesized on-demand via a multi-step enzymatic process initiated by trophic hormones (like ACTH or LH),

- Cytochrome P450 enzymes in the mitochondria and smooth ER of endocrine cells (adrenal cortex, gonads, placenta) convert cholesterol into active hormones. They are lipophilic and cannot be stored in vesicles and secreted immediately upon synthesis via diffusion through the plasma membrane.

- ## Synthesis, Storage, and Secretion:-

3. **Amine hormones:** are synthesized from amino acids like tyrosine or tryptophan via enzymatic decarboxylation. They are stored in high concentrations within cytoplasmic storage vesicles or granules (chromaffin granules in the adrenal medulla) until needed, often bound to proteins or ATP, protecting them from degradation.

- Upon stimulation (neural or hormonal), intracellular calcium levels increase, triggering the fusion of granules with the plasma membrane, After that these granules release their contents into the extracellular space or bloodstream via exocytosis.

- **Circulating hormones:** either free or bound to carrier/transport proteins w are synthesized by cells in the liver. most water soluble-hormone molecules circulate in plasma in a free form & don't attach to other molecules (protein).The plasma binding protein hormone complex allow lipophilic Hs to travel through blood stream. Only the free/ unbound hormones can leave the blood & get to the target cells. There is an equilibrium between unbound & bound fraction in the blood.

Functions of transport/carrier proteins:

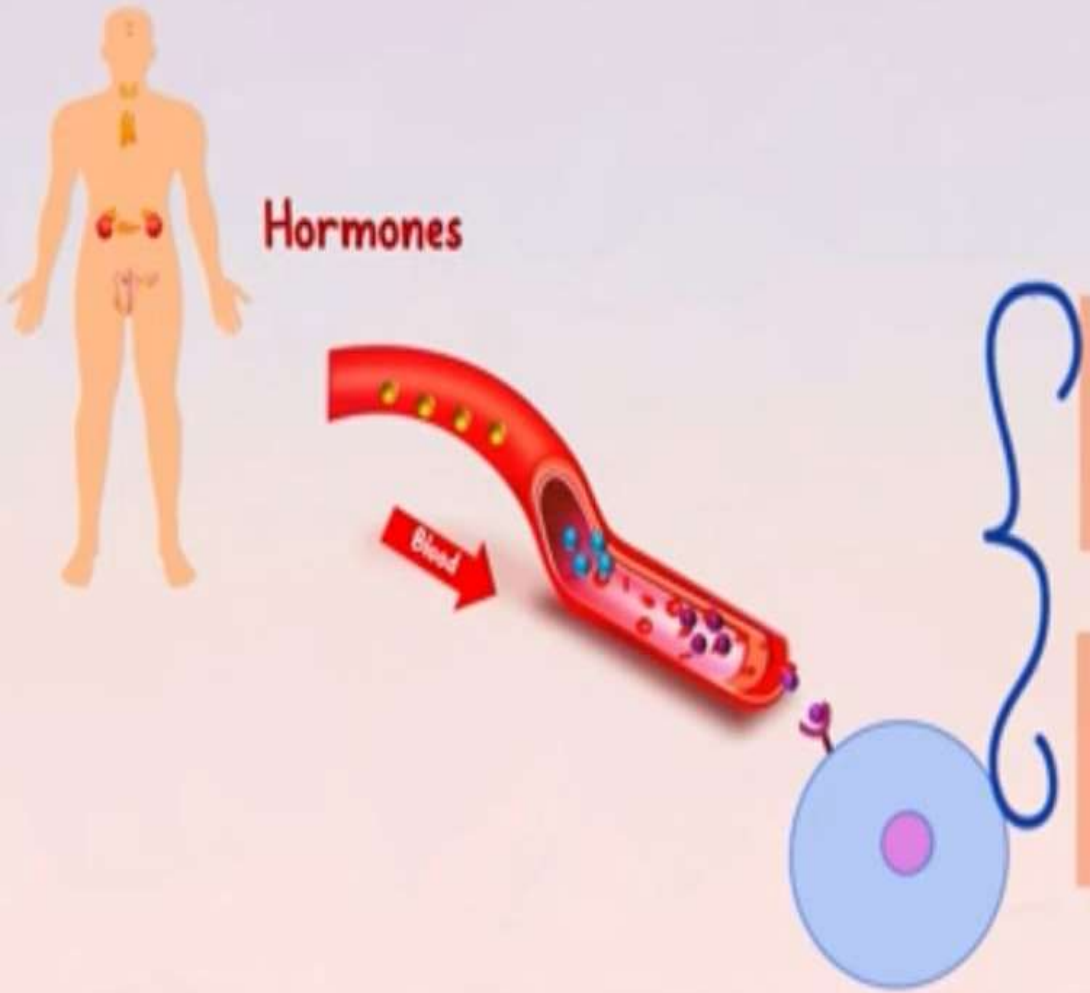
1. They make lipid soluble Hs temporarily soluble; thus increasing their solubility in the blood.

2. Retard the passage of small H molecules via the filtering mechanism in the Kidneys; thus slowing the rate of H loss in the urine.

3. The bound fraction acts as a reservoir; that helps to maintain steady plasma levels of the free Hs

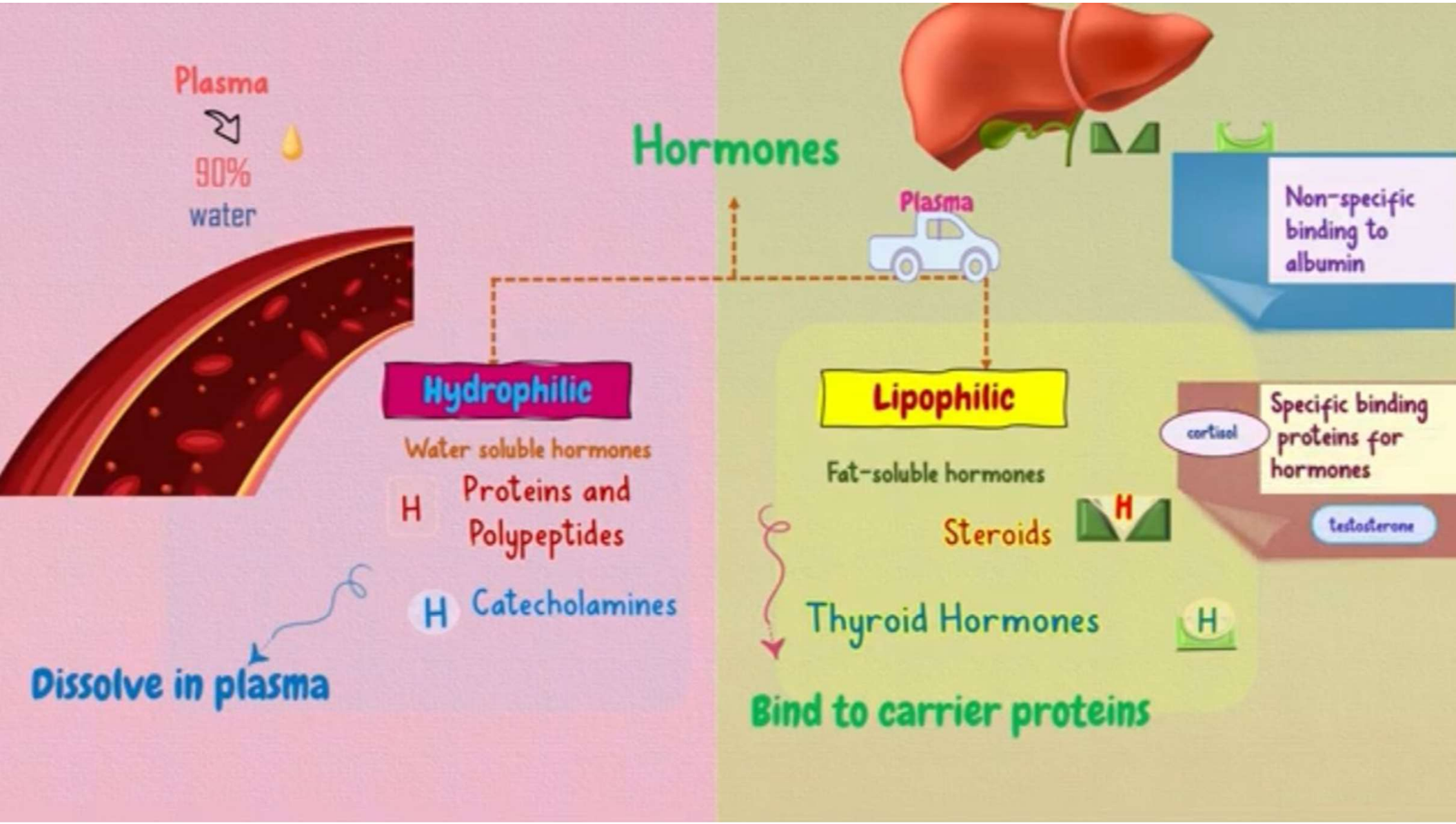
- **Free hormones:** last seconds to minutes: rapidly broken down by liver, kidney or plasma enzymes in blood.
- **Bound hormones :** inactive biologically. Become active when they are free H. last hours to days in blood. Effect at target cell can take seconds to days depending on mechanism and final effect, but hormone once bound to receptor is broken down quickly.

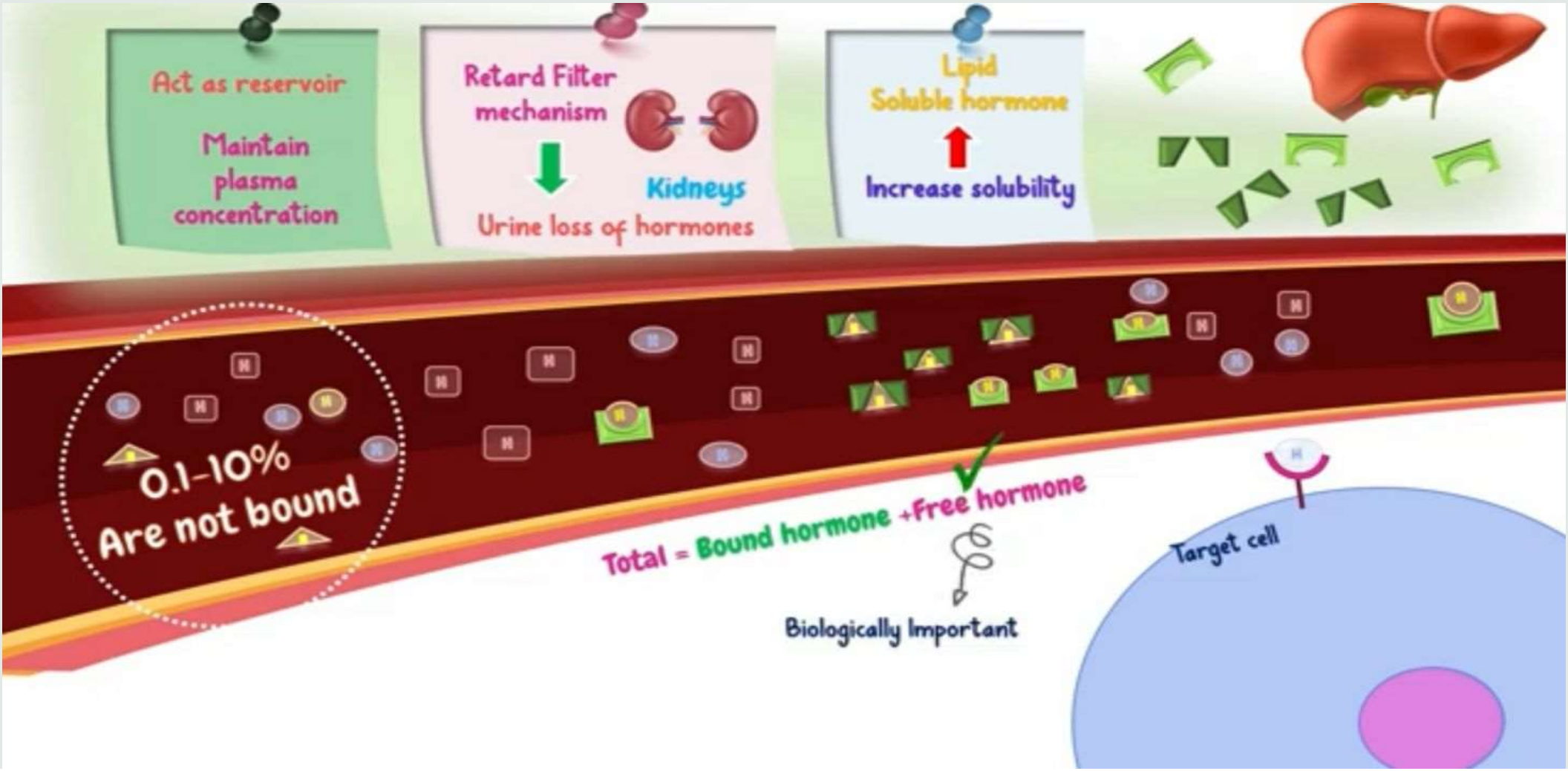
Hormonal Transport



Dissolve in plasma

Bound to plasma proteins


Plasma
90%
water
Hormones
Plasma
Non-specific binding to albumin
Hydrophilic
Water soluble hormones
H
Proteins and Polypeptides
H
Catecholamines
Dissolve in plasma
Lipophilic
Fat-soluble hormones
Steroids
H
Thyroid Hormones
H
Bind to carrier proteins
cortisol
Specific binding proteins for hormones
testosterone


Act as reservoir
Maintain plasma concentration
Retard Filter mechanism
Kidneys
Urine loss of hormones
Lipid Soluble hormone
Increase solubility
0.1-10% Are not bound
Total = Bound hormone +Free hormone
Biologically Important
Target cell

Hormone Receptors:-

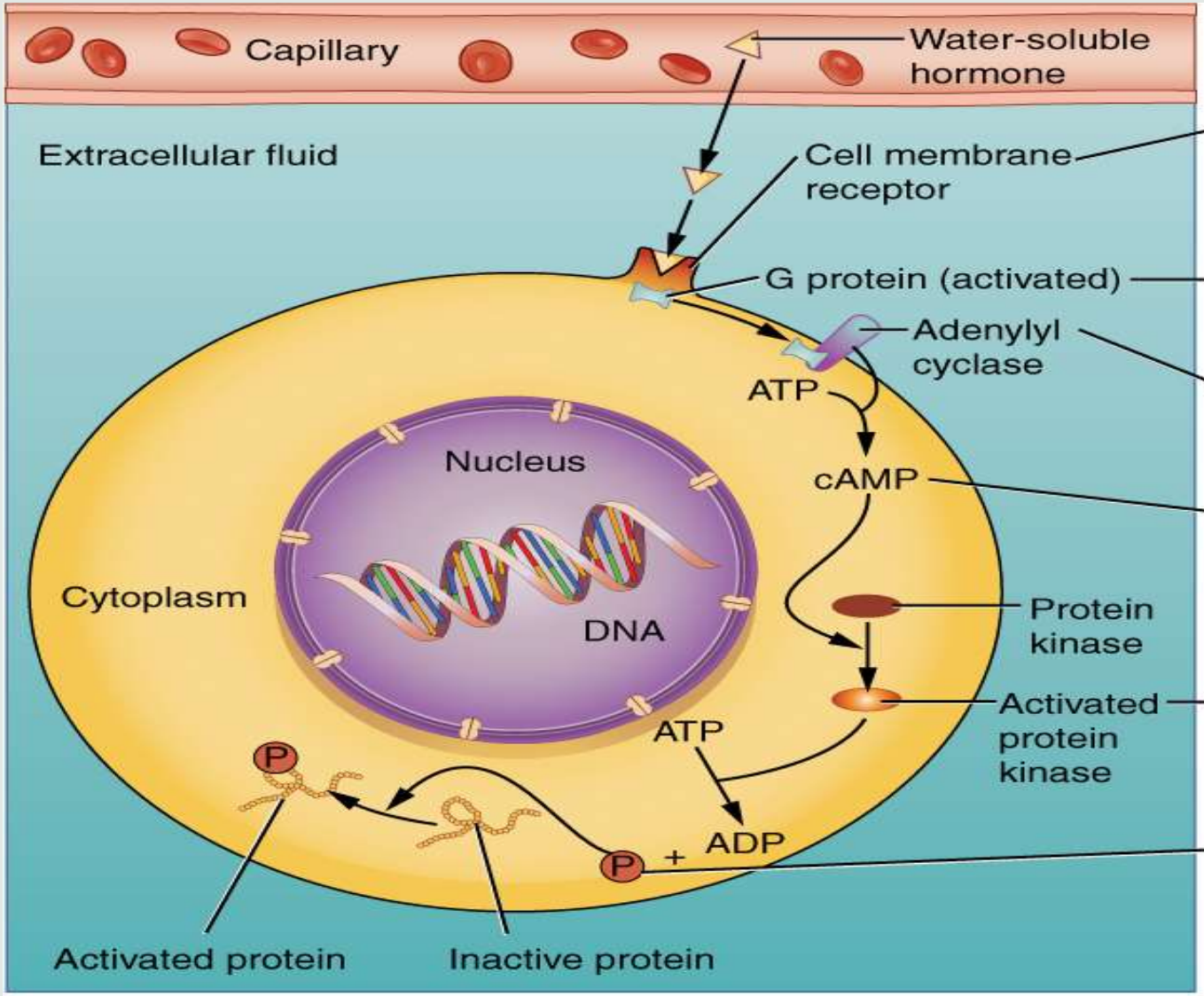
Located on plasma membrane or inside target cell.

1. Cell membrane hormone receptors:

They are specialized, high-affinity proteins embedded in the plasma membrane that bind to water-soluble hormones (like peptides and catecholamines, glycoprotein hormones, eicosanoids) which cannot cross the lipid bilayer. Binding to R (1st messenger) indirectly initiate a cascade events inside the cell via 2nd messengers (cAMP, Ca++) which act as activators, inhibitors, or cofactors for intracellular enzymes (enzymes catalyze reactions for cell changes).

Hormone stimulus effects in target cells:

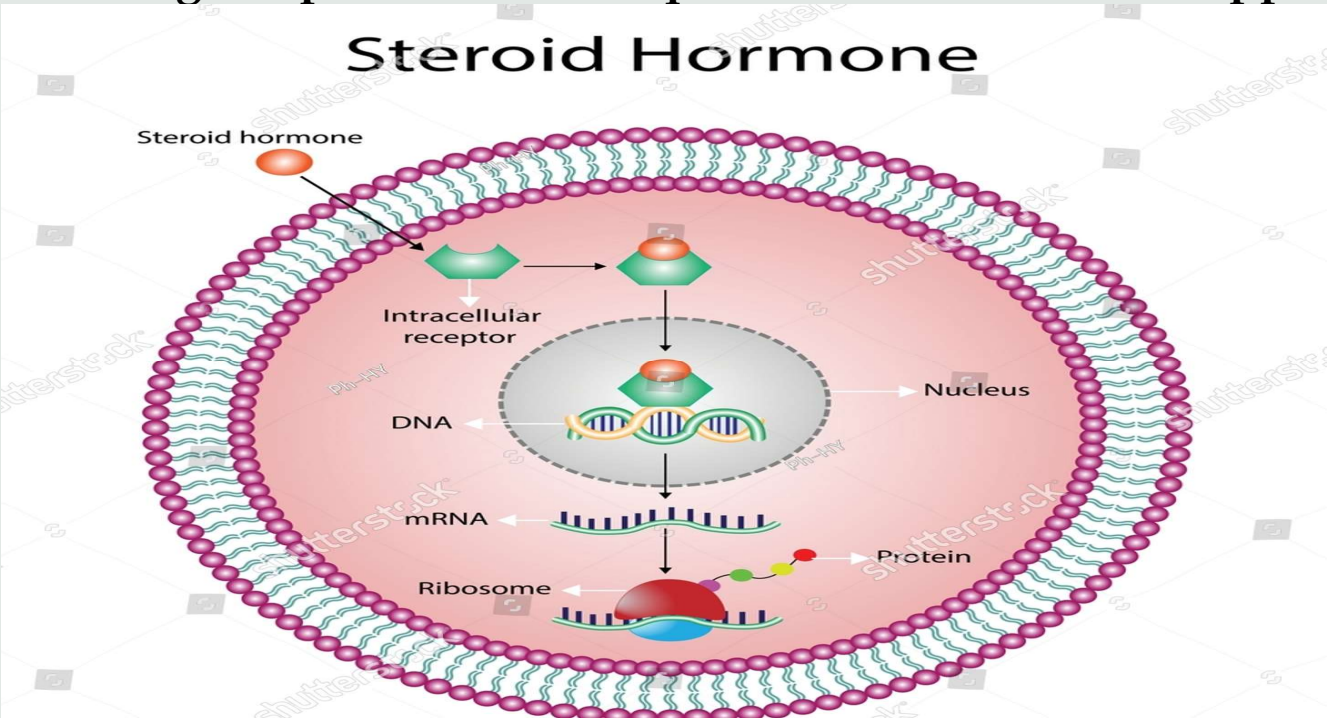
- **The extracellular signals are linked to intracellular 2nd messengers via the primary cell surface R (G protein) w is (a regulatory enzyme complex). Activation of G-proteins, often leading to the production of cyclic AMP (cAMP).**
  - **2nd messenger mechanism results in amplification of hormone signal: one hormone molecule binds one receptor but can result in millions of final products .**



1. Water-soluble hormones are membrane insoluble. They bind to membrane receptors.
2. The binding activates a G protein.
3. The activated G protein activates adenylyl cyclase.
4. Adenylyl cyclase catalyzes the conversion of ATP to cAMP, the secondary messenger in this pathway.
5. cAMP activates protein kinases.
6. Protein kinases phosphorylate proteins in the cytoplasm. This activates these proteins, allowing them to alter cell activity.

2. Intracellular hormone receptors:

- **They are protein-based receptors located within the cytoplasm (steroid receptors) or nucleus (thyroid receptors) that bind to lipophilic (lipid-soluble) hormones, such as steroid and thyroid hormones, which diffuse through the plasma membrane. Upon binding, these receptors act as transcription factors, directly regulating gene expression by binding to specific DNA sequences to enhance or suppress protein.**



Hormone Interactions: refer to how multiple hormones work together or against each other for regulation physiological processes & maintaining homeostasis, thus occur via binding to specific cellular receptors.

- The three primary types are:

1. Antagonistic (opposing effects): occurs when two or more hormones have opposite effects and compete for the same receptors.

Ex, **insulin and glucagon** have opposite effects on blood glucose levels.

Also, **Calcitonin and Parathyroid H** have opposite effects on blood calcium levels

2. Synergistic (amplified effect): occurs when two or more hormones work together to produce a stronger & amplified response than either hormone alone.

Ex, **insulin and growth hormone (GH)** work together to promote protein synthesis and cell growth.

Also, **GH and Thyroxine (T4)** work synergistically to promote growth, metabolism, and development particularly in children.

3. Permissive (one hormone enables another): occurs when one hormone enhances the responsiveness of a target cell to another hormone.

Ex, (**T4& epinephrine**) thyroid hormone increases the number of receptors for epinephrine in adipose tissue, which enhances lipolysis and thereby boosting epinephrine's impact on the body.

Also, (**TSH-RH and TSH**) both stimulates the thyroid gland to produce (T3 & T4)hormones for regulating metabolism.

Principles of endocrine function

- Endocrine control involves secretion of a hormone into the circulation.
- This messenger molecule binds to any cells which carry the specific receptors.
- A given substance may act both as a neurotransmitter and as a hormone at different sites.
- For example, nor-adrenaline (nor-epinephrine) is a postganglionic sympathetic neurotransmitter but is also released into the circulation from the adrenal medulla.

The release of endocrine substances is controlled in three ways:

Hormone production and release are primarily controlled by negative feedback. In negative feedback systems, a stimulus evokes the release of a substance; once the substance reaches a certain level, it sends a signal that stops further release of the substance. In this way, the concentration of hormones in blood is maintained within a narrow range.

There are three mechanisms by which endocrine glands are stimulated to synthesize and release hormones: humoral stimuli, hormonal stimuli, and neural stimuli.

The release of endocrine substances is controlled in three ways:

1- Humoral stimuli (control by a regulated solute):

"Humoral" is derived from the term "humor," that refers to body fluids such as blood. **A humoral stimulus** refers to the control of hormone release in response to changes in extracellular fluids such as blood or **the ion concentration in the blood**.

e.g. A rise in blood **glucose** levels induces the pancreatic release of **insulin**. Insulin reduces blood glucose levels, which signals the pancreas to stop producing insulin in **a negative feedback loop.**

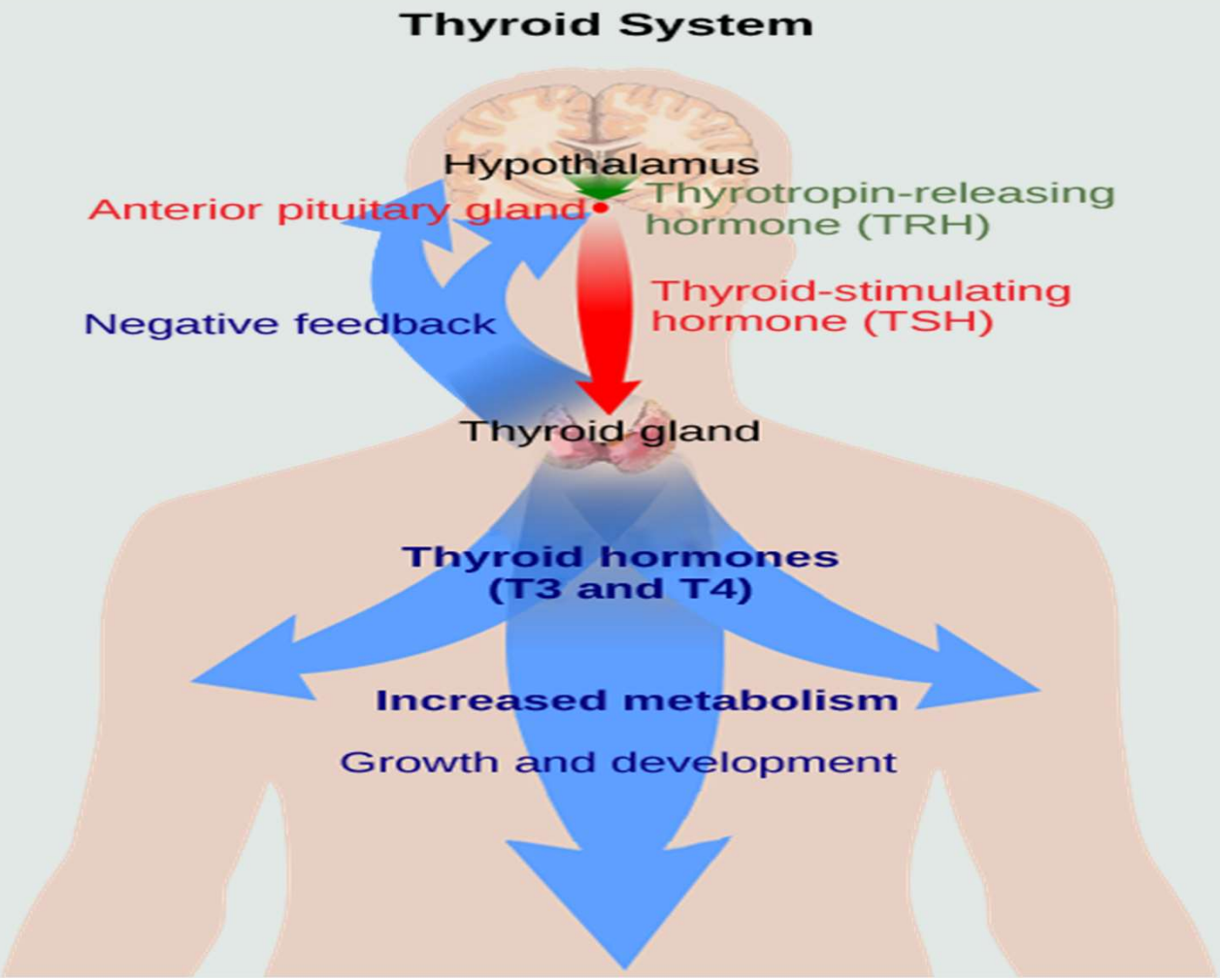
e.g. The cells which secrete **parathormone**, which acts to reduce the concentration of Ca2+ in the extracellular fluid, are themselves directly sensitive to [**Ca2+**].

2- Hormonal stimuli (control by a different hormone):

The release of a hormone in response to another hormone. A number of endocrine glands release hormones when stimulated by hormones released by other endocrine glands.

e.g. The hypothalamus produces hormones that stimulate the anterior portion of the pituitary gland. The anterior pituitary in turn releases hormones that regulate hormone production by other endocrine glands. The anterior pituitary releases the **thyroid-stimulating hormone (TSH)**, which then stimulates the thyroid gland to produce the hormones **T_3 and T_4**. As blood concentrations of T_3 and T_4 rise, they inhibit both the pituitary and the hypothalamus in a negative feedback loop.

The regulation of thyroid hormone secretion through a negative feedback loop



Feedback loops are a critical part of **homeostasis**, which is the tendency of organisms to maintain relatively stable internal environments. Maintaining homeostasis typically occurs through **negative feedback loops**.

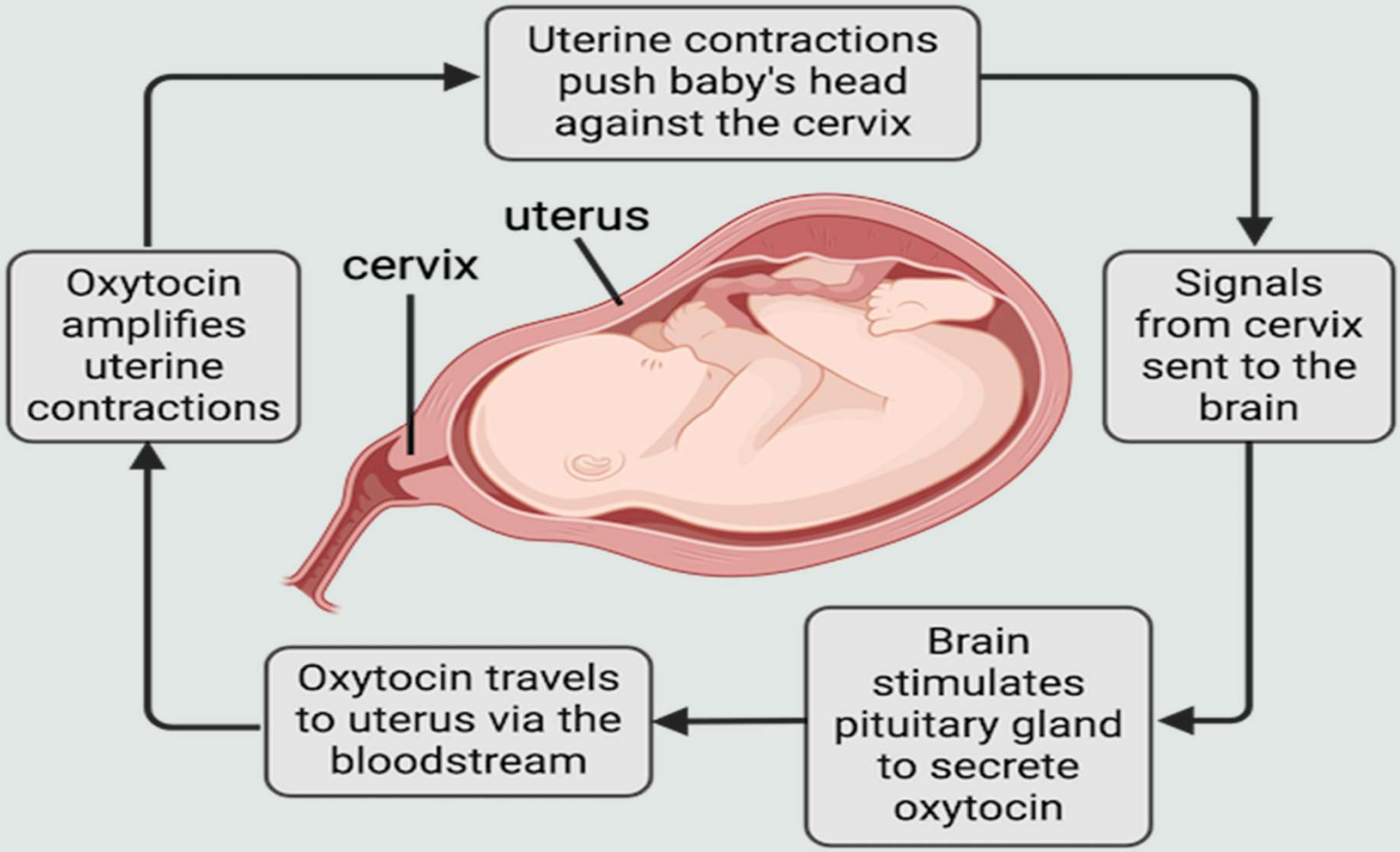
Some biological processes, however, require **positive feedback loops** to amplify a physiological response by causing a rapid, exponential rise in hormone levels rather than maintaining stability. **Positive feedback loops** are usually found in processes that need to be pushed to completion, not to be maintained.

Ex1. Oxytocin during Childbirth: Uterine contractions (stimulated by oxytocin) push the fetus against the cervix, sending signals to the brain to release more oxytocin, for further intensifying contractions. This cycle continues until the baby is delivered.

Ex2. Estrogen during the Menstrual Cycle: Rising levels of estrogen in the late follicular phase signaling the brain to trigger a massive, rapid surge of luteinizing hormone (LH) to induce ovulation.

Ex3. Oxytocin during Breastfeeding: The suckling action of a baby stimulates sensory nerves that trigger the pituitary gland to release more oxytocin, strengthening the milk-ejection.

The regulation of childbirth through a positive feedback loop



3- Neural stimuli (nervous control):

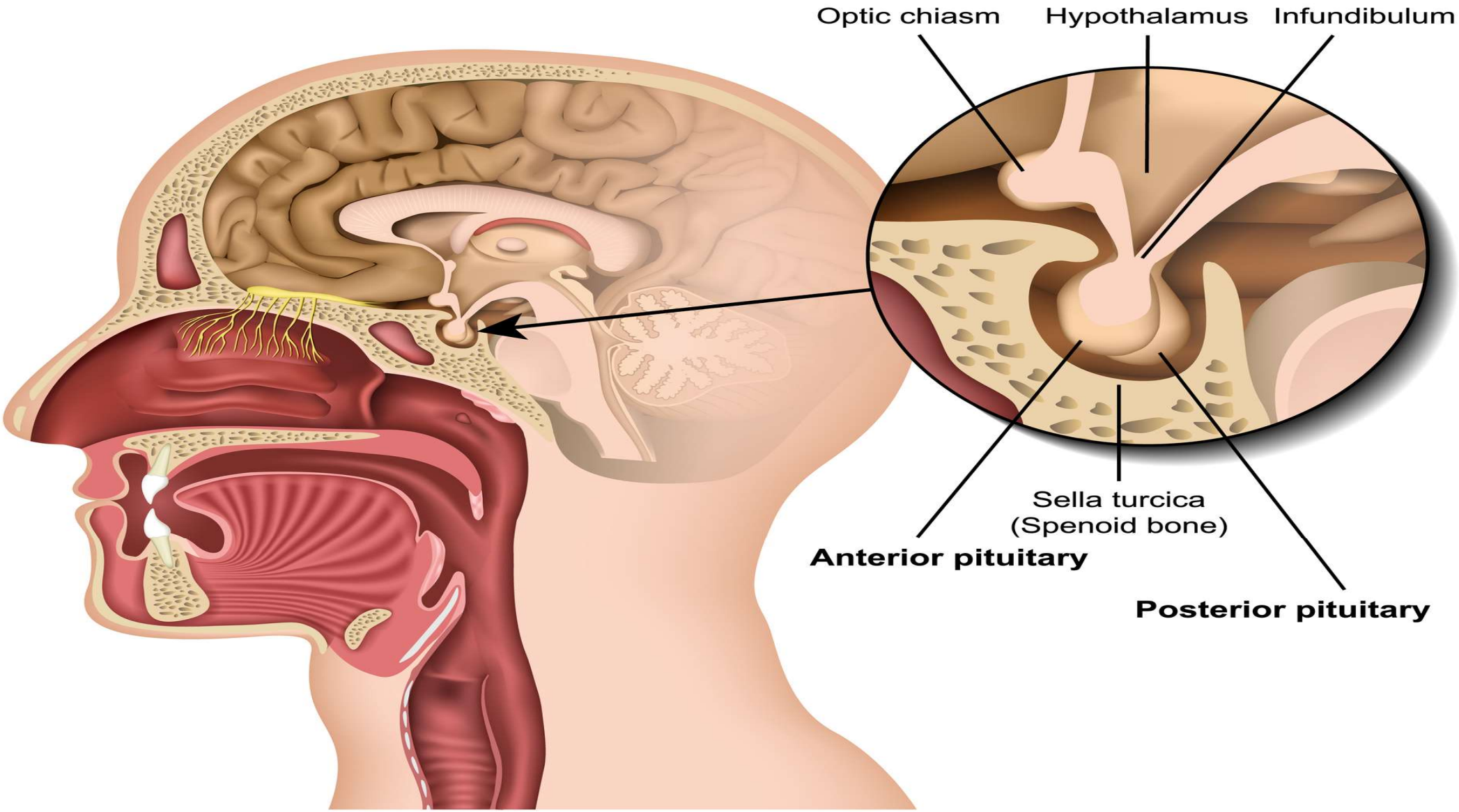
In some cases, the nervous system directly stimulates endocrine glands to release hormones, which is known as **neural stimuli**.

In a short-term stress response, the hormones **epinephrine and norepinephrine** are important for providing the surge of energy required for the body to respond. Here, neuronal signals from the sympathetic nervous system directly stimulates the adrenal medulla to release epinephrine and norepinephrine in response to stress.

- **The hypothalamus** plays a central role in linking various **endocrine systems with the nervous system**.

Hypothalamic and pituitary function

- The hypothalamus regulates both **autonomic nervous** activity and several aspects of **endocrine function**.
- The latter role is fulfilled through its links with **the pituitary gland**, which secretes a wide range of hormones. Some of these regulate animal physiology directly, while others control the activity of various endocrine glands around the body.
- **The hypothalamus** is located adjacent to the third ventricle in the forebrain and is connected by the hypophyseal stalk to the pituitary gland (hypophysis cerebri) immediately inferior to it.


Optic chiasm
Hypothalamus
Infundibulum
Sella turcica
(Spenoid bone)
Anterior pituitary
Posterior pituitary

The anterior pituitary (adenohypophysis)

- Is a classical endocrine gland containing three main types of hormone-secreting epithelial cells (**Acidophils**, **basophils** and **heterophyls cells or chromophobes).**
- **Acidophils (approx. 40% of cells)** , which are somatotropes and lactotropes,

Somatotropes: Most common, secrete Growth Hormone (GH).

Lactotropes: Secrete Prolactin (PRL).

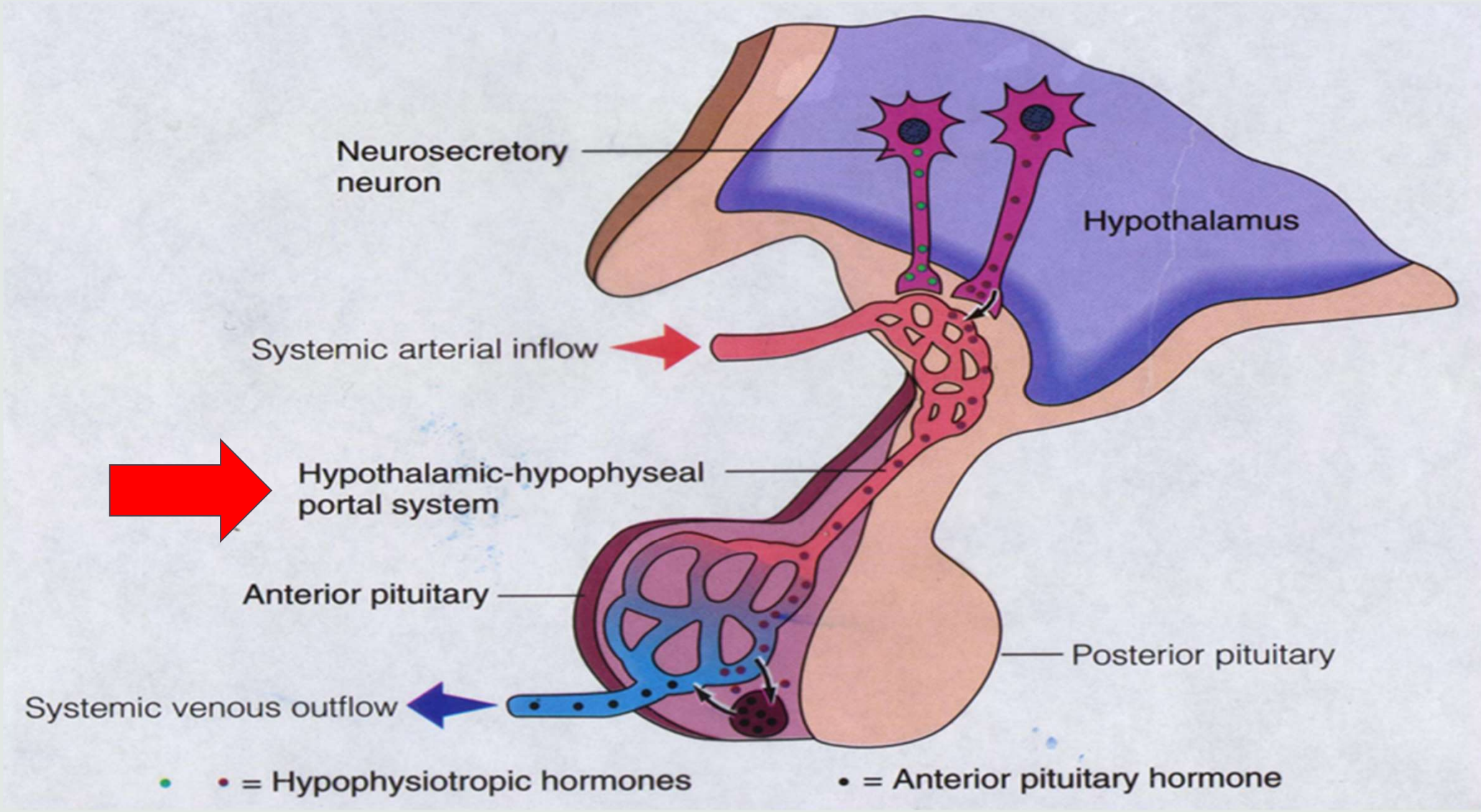
- **Basophils (approx. 10% of cells)** , consisting of corticotropes, thyrotropes, and gonadotropes

Corticotropes: Secrete Adrenocorticotrophic hormone (ACTH).

Thyrotropes: Secrete Thyroid-stimulating hormone (TSH).

Gonadotropes: Secrete Follicle-stimulating hormone (FSH) and Luteinizing hormone (LH)

- **Chromophobes (approx. 50% of cells):** considered resting or stem cells.


Neurosecretory neuron
Hypothalamus
Systemic arterial inflow
Hypothalamic-hypophyseal portal system
Anterior pituitary
Posterior pituitary
Systemic venous outflow
• = Hypophysiotropic hormones
• = Anterior pituitary hormone

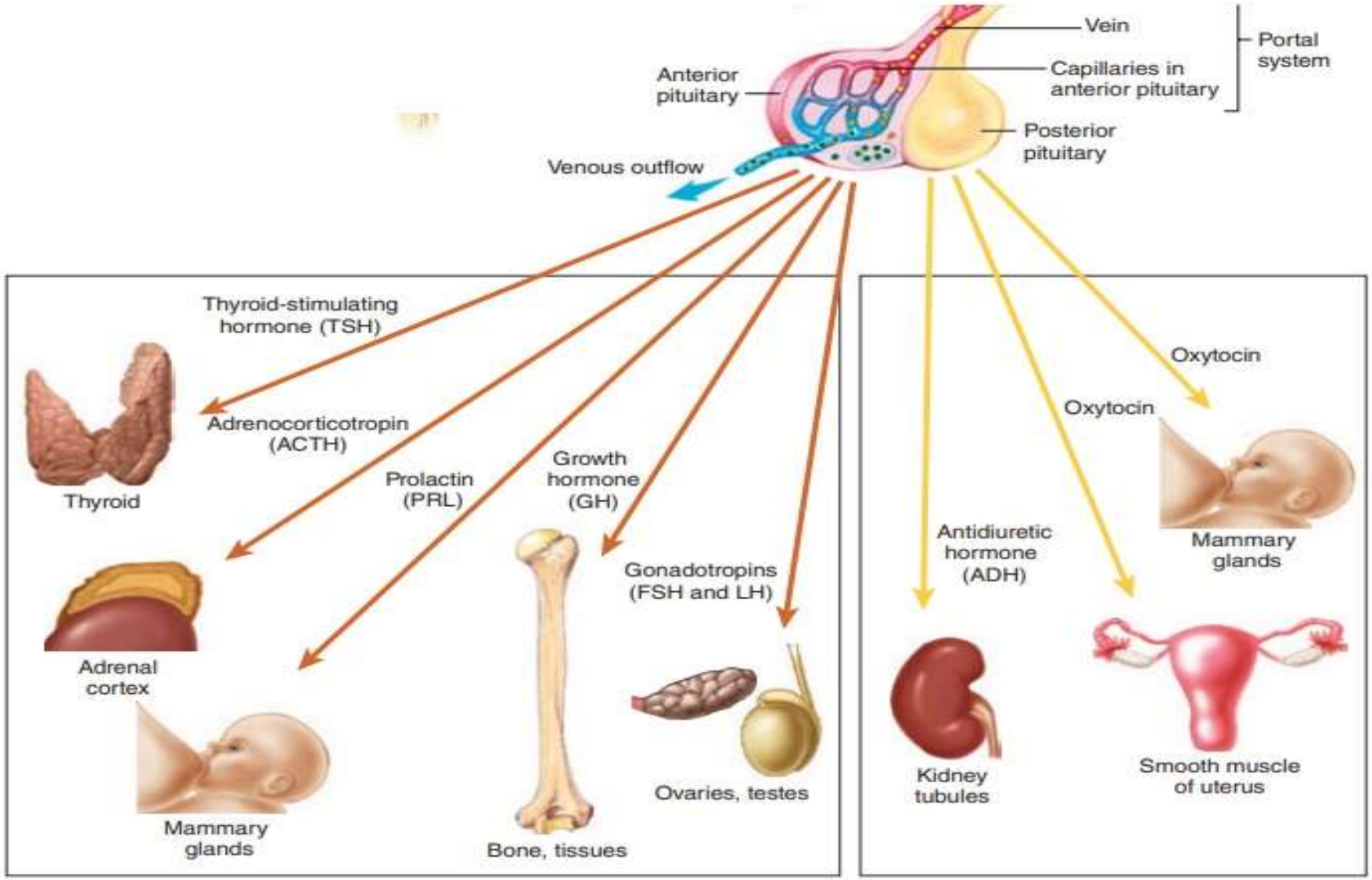
- The anterior pituitary receives its blood supply from a series of hypophyseal portal vessels which link it with the hypothalamus.
- The endocrine activity of the anterior pituitary is regulated by hormones secreted from the hypothalamus and transported in the hypophyseal portal blood.
- Each of these hormones selectively promotes or inhibits the secretion of a specific pituitary hormone and so they are referred to as releasing or inhibiting hormones, respectively.
- They are generally oligopeptides, although prolactin inhibiting hormone (PIH) is probably the transmitter substance dopamine.
- Hypothalamic secretion of these pituitary-regulating hormones is influenced by **neurological inputs** and **feedback control.**

Regulation of pituitary cyclicity:-

- If female rats 5 days of age are given a single dose (101.5 mg of androgen).
- 99% of them do not show normal estrous cycle when mature sexually.
- Injections before or after the 5th day of age are ineffective.
- So, the hypothalamus is normally "set" to be cyclic and that in order to become acyclic it must receive androgen of testicular origin produced by males own testis at day 5 of life.

Q: Discuss the physiological basis of Freemartins.

- A freemartin is an infertile female bovine (or other ruminant) born twin to a male, rendered sterile by vascular anastomosis (shared blood vessels) between their placentas during early gestation around day 30 of gestation in cattle. As male gonadal development occurs earlier than females.
- The male twin's testes produce AMH and testosterone, which travel to the female fetus and cause the regression of the female Müllerian ducts (which form the uterus, oviducts, and vagina). Also, causes the female fetus to have a mixed population of cells, 80% to 95% or higher (XX/XY) in the blood, leading to incomplete sex development.



The pituitary gland.

- **The anterior pituitary** secretes six peptide hormones.
- These include hormones controlling the endocrine functions of the adrenal cortex, the thyroid and the gonads, as well as prolactin and GH.

1. Adrenocorticotrophic hormone (ACTH) :

Is secreted in response to corticotrophin-releasing hormone (CRH) from the hypothalamus. Its action is to stimulate the release of glucocorticoids from the adrenal cortex.

A number of other peptides including the endogenous opiate β-endorphin (inducing analgesia, and stress reduction). Derived from the precursor proopiomelanocortin (POMC), it is produced in the anterior pituitary alongside melanocyte-stimulating hormone (MSH) and ACTH.

2. Thyroid-stimulating hormone (TSH) or thyrotrophin.

- Is secreted in response to thyrotrophin-releasing hormone (TRH) from the hypothalamus. It stimulates thyroid secretion.

3. Follicle-stimulating hormone (FSH) and luteinizing hormone (LH)

- Are secreted in response to gonadotrophin releasing neurons, gonadotrophin releasing hormone (GnRH) from the hypothalamus. These gonadotrophins stimulate the male and female gonads .

4. Growth hormone

5. Prolactin hormone

thank
you